

Audio Deep Learning

Course Project

Assignment overview and motivation

- *Type: Signal Deep Learning full-scale project, where students can apply the complete range of instruments covered in the course.*
- *Goal: Solve a real-world signal deep learning problem using methods from the course.*
- *Grading: 25 points (20 for implementation and results, 5 for presentation/report).*
- *Timeline: Final submission - July 19. Presentation - July 23*

Solving a general or formally defined Audio Deep Learning problem (for example, in the style of a Kaggle competition, research project, or industrial audio task) will help you gain the following skills:

- *Perform the full cycle of the Machine Learning development process — from EDA to final model or ensemble selection.*
- *Experiment with a wide range of architectures and methods such as CNNs, Transformers, Self-Supervised Learning models, sequence modeling approaches, generative models, and hybrid architectures combining signal processing and deep learning.*
- *Design and evaluate audio processing pipelines, including feature extraction, augmentation, representation learning, training, inference, and post-processing stages.*
- *Define an experiment grid based on domain knowledge and inductive bias, and select the final solution considering both objective performance metrics and practical constraints such as latency, robustness, and computational efficiency.*
- *Develop the ability to transfer methods across different audio domains, including speech processing, speaker analysis, environmental sound recognition, bioacoustics, and other signal-related tasks.*

What you will deliver

- *Code (on GitHub):*
 - *Demonstrate your EDA.*
 - *Show your experiment management and tracking system.*
 - *Ensure your solution is fully reproducible.*
 - *Include functionality to perform inference on new data.*
- *Report*
 - *Describe the task and its main challenges.*

- Highlight the most important data insights.
- Emphasize the main challenges and findings during modeling.
- Present an ablation study.
- Compare different approaches based on objective metrics (and subjective ones if needed).
- Draw conclusions on the final model and ensemble.
- Discuss possible limitations of your approach (both data- and task-related) and outline future work.
- Presentation
 - Highlight the main aspects of your report.
 - Engage the audience by focusing on the key findings and insights.

Important Notes:

You can choose 3 possible approaches to the project:

1. Solve the task from scratch, including data aggregation and benchmark construction till building models and deploying them. Example: Build an Automatic Speech Recognition (ASR) system for Ukrainian emergency service calls. Collect and curate audio recordings, prepare transcriptions, establish train/validation/test splits, fine-tune a modern ASR model, evaluate recognition quality using WER, and deploy an inference pipeline for real-time or batch transcription.
2. Dive deep into one/two approach architecture or method. Example: Propose and evaluate a new architectural building block or training methodology for speaker diarization. For instance, modify an existing segmentation-and-clustering pipeline with a custom speaker-aware temporal aggregation block, overlap-aware loss, or improved embedding fusion strategy. Compare the proposed method against a strong baseline, perform ablation studies, and analyze whether the new component improves DER, robustness to overlapping speech, or generalization across recording conditions.
3. Explore a wide variety of methods. Example: Solve an environmental sound classification task by comparing handcrafted acoustic features with classical machine learning models, CNNs trained on spectrograms, transformer-based architectures, self-supervised audio representations, and multimodal approaches incorporating metadata. Optionally combine the best-performing methods into an ensemble system.

Taking one basic approach and applying only it will result in a very small number of points for coursework

We estimate the amount of effort you invest in this task as X2-X3 from the basic Assignment

Choosing a topic:

- Kaggle competitions - you can take up one of the Kaggle Competitions. **Each competition can be selected by a maximum of 2 teams. It is forbidden to take the same competition (or competition from the same annual series, like BirdCLEF series) that you have taken for previous Machine Learning, Deep Learning, Computer Vision, and Natural Language Processing courses**
 - <https://www.kaggle.com/competitions/birdclef-2026/leaderboard>
 - <https://www.kaggle.com/competitions/bengaliai-speech>
 - <https://www.kaggle.com/competitions/ariel-data-challenge-2025>
 - <https://www.kaggle.com/competitions/waveform-inversion>
 - <https://www.kaggle.com/competitions/child-mind-institute-problematic-internet-use>
 - <https://www.kaggle.com/competitions/g2net-detecting-continuous-gravitational-waves>

- and others (**BUT MAKE SURE TO HAVE THEM APPROVED BY THE COURSE TEACHERS.**)
- Research projects. **Each competition can be selected by a maximum of 2 teams.** For example:
 - Improve the Dataset Preparation methodology based on some algorithmic methods and NNs. The original was introduced in [Andrii's master's thesis](#), Section 5.1 and Figure 5.1. Improvements may include
 - Develop drone detection and tracking algorithms based on data from [Acoustic Drone Package Delivery Detection paper](#)
 - Domain Adaptation for Bird Audio Classification. Investigate how bird sound recognition systems can be adapted to changing environmental conditions, geographical regions, urban environments, migration patterns, or newly observed species. Possible directions include transfer learning, domain adaptation techniques, test-time adaptation, and semi-supervised learning using external bird audio repositories. - write Volodymyr Sydorskyi (vsydorskyi@kse.org.ua) for more details
 - Self-Supervised Learning for Bioacoustics. Explore self-supervised learning approaches for bird audio analysis. Potential topics include training SSL models on large-scale unlabeled bird recordings, evaluating learned representations on downstream classification tasks, studying clustering properties of learned embeddings, or incorporating semantic information into representation learning. - write Volodymyr Sydorskyi (vsydorskyi@kse.org.ua) for more details
 - Generative Models for Bird Sounds. Develop specialized encoder-decoder, autoregressive, diffusion-based, or language-modeling approaches for bird sound generation. Possible objectives include realistic bird vocalization synthesis, species-conditioned generation, controllable generation, or augmentation of limited datasets through synthetic audio generation. - write Volodymyr Sydorskyi (vsydorskyi@kse.org.ua) for more details
 - Bird Audio Denoising and Source Separation. Investigate methods for separating bird vocalizations from environmental noise. Potential directions include isolating a target bird species from background sounds, separating multiple simultaneously vocalizing birds into individual tracks, removing non-biological noise sources such as wind and rain, or building source-separation systems that improve downstream bird classification performance. - write Volodymyr Sydorskyi (vsydorskyi@kse.org.ua) for more details
 - Ukrainian ASR for specific use-cases - write Volodymyr Sydorskyi (vsydorskyi@kse.org.ua) for more details
 - BirdCall segmentation competition - will be announced soon
 - Ukrainian Automatic Speech Translation. In this task, you can go beyond classic ASR and implement a speech translation system. Your goal is to train a model to take source-language audio as input and generate Ukrainian text as output (e.g., French audio transcribed into Ukrainian text). The goal is to release a new UA AST dataset and SOTA model on HF and contribute to the development of Ukrainian Audio ML. Write Danil Brovko (dbrovko@kse.org.ua) for more details.
- Future work based on one of the latest papers **in the domain of Audio and Signal Processing** from the next conferences:
 - [NeurIPS](#)
 - [ICML](#)
 - [ICLR](#)
 - [ICASSP](#)
 - [Interspeech](#)
- You are welcome to suggest your own ideas – **BUT MAKE SURE TO HAVE THEM APPROVED BY THE COURSE TEACHERS.**

Grading Policy

- *Type: Individual or group (max 5 students).*
- *Goal: Solve a real-world computer vision problem using methods from the course.*
- *Timeline: Final submission - July 19. Presentation - July 23*
- *Grading: 25 points (20 for implementation and results, 5 for presentation/report).*

Criterion	Points
<i>Problem Definition & Design</i>	<i>5</i>
<i>Implementation Correctness</i>	<i>10</i>
<i>Experimentation & Results</i>	<i>5</i>
<i>Final Presentation & Report</i>	<i>5</i>